

TWO-DIMENSIONAL DYNAMIC MODELLING OF HUMAN KNEE JOINT

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Abstract—A mathematical dynamic model of the two-dimensional representation of the knee joint is presented. The profiles of the joint surfaces are determined from X-ray films and they are represented by polynomials. The joint ligaments are modelled as nonlinear elastic springs of realistic stiffness properties. Nonlinear equations of motion coupled with nonlinear constraint conditions are solved numerically. Time derivatives are approximated by Newmark difference formulae and the resulting nonlinear algebraic equations are solved employing the Newton–Raphson iteration scheme. Several dynamic loads are applied to the center of mass of the tibia and the ensuing motion is investigated. Numerical results on ligament forces, contact point locations between femur and tibia, and the orientation of tibia relative to femur are presented. The results are shown to be consistent with the anatomy of the knee joint.

INTRODUCTION

Although the studies on the biomechanics analysis of the knee joint have a long history, those studies which are essentially on mathematical modelling of this largest and, apparently, most complicated joint in the body are few. Crowninshield *et al.* (1976) felt justified in stating that there were, at that time, no analytical models of the knee available which permit the prediction of the response of the joint to either external forces or displacements. In their analytical model, on the other hand, only the quasi-static response of the joint was studied and the overall joint stiffness was obtained as a function of the flexion angle. The method used by Crowninshield *et al.* (1976) is the so-called inverse method in which the ligament forces caused by a specified set of translations and rotations in specified directions are determined by comparing the geometries of the initial and displaced configurations of the knee joint. In this method the externally applied displacements do not need to be continuous; however, the discrete values used are to be realistic. In Crowninshield *et al.* (1976) these values were based on experimental data available in the literature. The purpose of the work was to obtain the stiffness of the joint which, in turn, required the calculation of the ligament lengths at various knee configurations. Thus it was not necessary to consider the contribution of the curved joint surfaces to the overall mechanical behavior of the knee. Moreover, the external forces required for equilibrium were not determined either.

Improvements to the quasi-static model discussed above have been provided recently by Wismans *et al.* (1980), in which a three-dimensional analytical model of the femoro-tibial joint is presented. The study considers not only the geometry but also the static equilibrium of the system. The three-dimensional curved geometry of the joint surfaces are included in the model. Ligaments are modelled as non-linear elastic springs. The solution method employed by Wismans *et al.* (1980) is also a quasi-static, inverse method. The flexion–extension motion is simulated by prescribing several flexion–extension angles. The dependent variables of the problem, including the ligament forces, are determined from the equilibrium equations and the geometric compatibility conditions. However, for nonlinear problems of this kind it is known that there can exist more than one equilibrium configuration for a given flexion–extension angle unless an external force is also specified. Accordingly, in the inverse method utilized by Wismans *et al.* (1980) it is necessary to specify the external force required for the preferred equilibrium configuration. Such an approach is applicable in quasi-static analysis only. In dynamic analysis, on the other hand, the equilibrium configuration preferred by the system is the unknown and the mathematical analysis itself is to provide that equilibrium configuration.

In addition to the two investigations discussed above, one can cite the related works by Menschik (1974) and Huson (1974, 1976) in which four-bar mechanisms were used to model the knee joint. In such models, the influence of the various components of the joint on the response cannot be studied. Biomechanics

of the knee joint have also been investigated via a nonlinear finite element model (Andriacchi *et al.*, 1977). Both material and geometric nonlinearities are included in the three-dimensional model of the joint and its associated components. Numerical predictions are consistent with experimental observations. The models of Andriacchi *et al.* (1977) and Wismans *et al.* (1980) are essentially the same in the sense that both deal with the quasi-static response of the knee joint.

As seen from the discussion presented above, mathematical modelling of the knee joint has not yet reached the stage it deserves. It is interesting to note that a *biodynamic* model of the knee joint is, to the authors' knowledge, not yet available in the literature. It is more appropriate to study via a dynamic model the response of the joint to dynamically applied loads. The artificial restrictions of the quasi-static inverse method, such as the necessity to specify the preferred configuration, can be eliminated if the dynamics of the problem is incorporated into the model. The most general and realistic dynamic model of the knee joint should, of course, be three dimensional, however, it is believed that a simpler two dimensional model should be helpful and rewarding in an understanding of the problems that may be associated with the three-dimensional dynamic model. Moreover, by restricting the motion to two dimensions it will hopefully be possible to dwell upon the essentials of the problem which will, in turn, lead to a sound development of the more realistic three-dimensional model.

The purpose of the present work is to present the development of a two-dimensional biodynamic model of the knee joint. Such a biodynamic model can be a part of a multi-segmented total-human-body model which is used in predicting biodynamic response of the human body subjected to expected or unexpected

external load conditions. These multi-segmented models play a significant role in both vehicle crash victim studies and aerospace-related applications. Our present model includes both geometric and material nonlinearities. The highly nonlinear set of governing equations are solved numerically and the numerical procedure is presented in sufficient detail. The effect of the duration and the shape of the dynamically applied loads on the response of the joint is discussed. Although the mathematical development to be presented is general, the numerical results will be for the human knee. Detailed discussions of various anatomical and functional aspects of the human knee joint can be found in Gray (1973), Engin and Korde (1974) and Engin (1978).

FORMULATION

In the present work the knee joint will be modelled by two rigid bodies connected by nonlinear elastic elements simulating the ligaments. It is assumed that the femur is rigidly fixed and the tibia is undergoing a general plane motion relative to the femur. Due to the two-dimensional nature of the model proper components of the ligament forces in the plane of motion are considered. The coefficient of friction between the articulating surfaces, owing to the presence of the synovial fluid, is known to be very low (Radin and Paul, 1972). Accordingly, the friction force between the femoral and tibial articulating surfaces will be neglected.

Geometry and kinematic conditions

The position of the moving segment (tibia) relative to the fixed segment (femur) is described by two coordinate systems as shown in Fig. 1. The origin of the

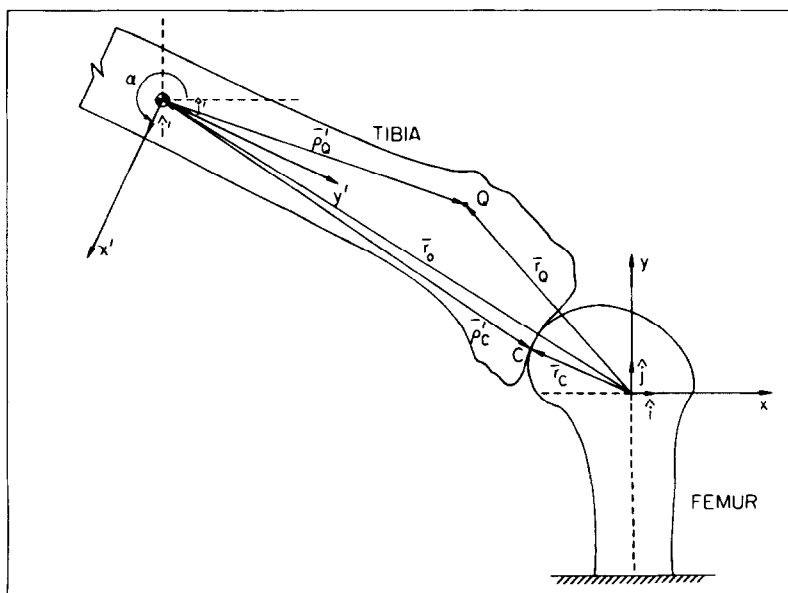


Fig. 1. Coordinate systems locations and relative positions of the tibia and femur are shown for the two-dimensional dynamic model of the knee joint.

moving coordinate system (x', y') coincides with the center of mass of the tibia, the y' -axis being directed along the longitudinal axis of the tibia. The inertial coordinate system (x, y) is attached to the fixed femur with the x -axis directed along the posterior-anterior direction and the y -axis coinciding with the femoral longitudinal axis. The locations of the origins of these coordinate systems shown in Fig. 1 are 4.01 and 21.34 cm away from the intersection points of the y and y' axes with the articulating surfaces. The rotation of the moving (x', y') coordinate system with respect to the fixed (x, y) system is denoted by α . The position vector of the origin of the (x', y') system in the fixed system is given by

$$\bar{r}_o = x_o \hat{i} + y_o \hat{j}, \quad (1)$$

where $\hat{i}$ and $\hat{j}$ are the unit vectors along the x and y directions, respectively. The base vectors along the moving x' and y' directions will be denoted by $\hat{i}'$ and $\hat{j}'$, respectively. Let $\bar{p}'_Q$ be the position vector of an arbitrary point Q on the tibia in the base $(\hat{i}', \hat{j}')$. Let $\bar{r}_Q$ be the position vector of the same point in the base $(\hat{i}, \hat{j})$. That is,

$$\bar{p}'_Q = x'_Q \hat{i}' + y'_Q \hat{j}', \quad (2)$$

$$\bar{r}_Q = x_Q \hat{i} + y_Q \hat{j}. \quad (3)$$

It can be shown that

$$\{r_Q\} = \{r_o\} + [T] \{p'_Q\} \quad (4)$$

where $[T]$ is an orthogonal transformation matrix whose elements are

$$[T] = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}. \quad (5)$$

Assuming rigid body contact between the tibia and the femur at point C as shown in Fig. 1, let us represent contact surfaces by smooth mathematical functions of the following form:

$$y = f_1(x) \quad (6a)$$

$$y' = f_2(x'). \quad (6b)$$

As implied, (6a) and (6b) represent the femoral and tibial articulating surfaces, respectively.

The position vectors of the contact point C in the base $(\hat{i}, \hat{j})$ is denoted by $\bar{r}_c = x_c \hat{i} + f_1(x_c) \hat{j}$ and the corresponding one in the base $(\hat{i}', \hat{j}')$ is $\bar{p}'_c = x'_c \hat{i}' + f_2(x'_c) \hat{j}'$. From Fig. 1, it is seen that

$$\{r_c\} = \{r_o\} + [T] \{p'_c\}. \quad (7)$$

This is the geometric compatibility condition for the problem under consideration. Moreover, the normals to the surfaces of the tibia and the femur at the point of contact must be colinear. Let $\hat{n}_1$ be the unit normal to the femoral surface directed toward its center of curvature. Let $\hat{n}_2$ be the unit normal to the tibial surface directed toward the center of curvature of the tibial surface. A lengthy but straightforward analysis yields:

$$\hat{n}_1 = \frac{(d^2 f_1 / dx^2)}{|d^2 f_1 / dx^2|} \left[1 + \left(\frac{df_1}{dx} \right)^2 \right]^{-1/2} \left[- \left(\frac{df_1}{dx} \right) \hat{i} + \hat{j} \right], \quad (8a)$$

$$\hat{n}_2 = \frac{(d^2 f_2 / dx'^2)}{|d^2 f_2 / dx'^2|} \left[1 + \left(\frac{df_2}{dx'} \right)^2 \right]^{-1/2} \left[- \left(\cos \alpha \frac{df_2}{dx'} + \sin \alpha \right) \hat{i} + \left(\cos \alpha - \sin \alpha \frac{df_2}{dx'} \right) \hat{j} \right], \quad (8b)$$

where $\hat{n}_2$ has already been transformed to the base $(\hat{i}, \hat{j})$. The colinearity of the normals at point C requires that the cross product of $\hat{n}_1$ and $\hat{n}_2$ be zero, i.e.

$$\hat{n}_1 \times \hat{n}_2 = \hat{0} \quad \text{at} \quad x = x_c, x' = x'_c. \quad (9)$$

The contact condition, thus takes the following final form:

$$\begin{aligned} & \sin \alpha \left[1 + \left(\frac{df_1}{dx} \right)_{x=x_c} \cdot \left(\frac{df_2}{dx'} \right)_{x'=x'_c} \right] \\ & - \cos \alpha \left[\left(\frac{df_1}{dx} \right)_{x=x_c} - \left(\frac{df_2}{dx'} \right)_{x'=x'_c} \right] = 0. \end{aligned} \quad (10)$$

The two-dimensional profiles in the plane of motion of the femoral and tibial articulating surfaces are obtained from the X-ray of a human knee joint. The coordinates of a number of points on these profiles are measured using a two-dimensional sonic digitizing technique. A polynomial equation of degree n ($n > 1$) is used as an approximate mathematical representation of the profile under consideration. The coefficients of polynomial are determined using a subroutine which is capable of determining, by means of statistical tests, whether the set of given data points are linear or nonlinear. If the data are nonlinear at the 95% confidence level, then the routine finds the lowest degree polynomial which adequately represents the data. The calculation procedure is to compare the standard deviation of the model of degree n with that of $n+1$ st degree model. If there is no significant difference at the 95% confidence level, then the polynomial of degree n is accepted as the lowest degree polynomial which adequately represents the data. In the present study, the method described above yielded the following equations for the femoral and tibial profiles, respectively:

$$f_1(x) = 0.04014 - 0.247621x - 6.889185x^2 - 270.4456x^3 - 8589.942x^4, \quad (11a)$$

$$f_2(x') = 0.213373 - 0.0456051x' + 1.073446x'^2. \quad (11b)$$

These are the functions used in the contact condition (10).

Ligament and contact forces

During its motion the tibia is subjected to the ligament forces, contact forces and the externally applied forces and moments. The contact force and the ligament forces are the unknowns of the problem and the external forces and moments will be specified.

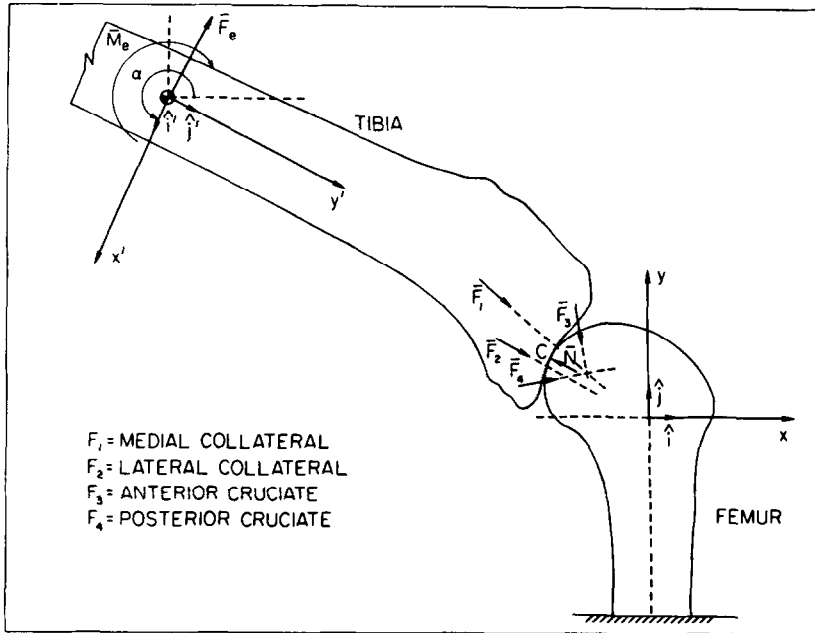


Fig. 2. Forces acting on the moving tibia are shown for the two-dimensional model of the knee joint.

These forces are discussed in some detail in the following paragraphs.

Only four major ligaments of the knee joint will be considered in the present work although consideration of any other ligament presents no difficulty. These ligaments are the lateral collateral (LC), the medial collateral (MC), the anterior cruciate (AC), and the posterior cruciate (PC) (Fig. 2). The ligaments are modelled as nonlinear elastic springs. To be precise, the following force-elongation relationship is assumed for each ligament:

$$F_m = K_m(L_m - l_m)^2 \quad \text{for } L_m > l_m, \quad (12a)$$

in which K_m is the spring constant, L_m and l_m are, respectively, the current and initial lengths of the ligament m . The tensile force in the m th ligament, thus, designated by F_m . It is assumed that the ligaments cannot carry any compressive force; accordingly;

$$F_m = 0 \quad \text{for } L_m < l_m. \quad (12b)$$

The stiffness values, K_m , are estimated according to the data available in the literature (Trent *et al.*, 1976; Kennedy *et al.*, 1976), and the values used in the present work are given in Table 1. These are the same values as those used in Wismans *et al.* (1980). Initial strains in the ligaments are taken as zero since, at present, there is no accurate data available on these strains as a function of

flexion angle. Zero strain condition for the ligaments can be partially justified if an appropriate starting flexion angle under no external load is chosen.

Let $(\bar{\rho}'_m)_t$ be the position vector in the base $(\hat{i}, \hat{j})$ of the insertion point of the ligament m in the moving tibia. The position vector of the origin point of the same ligament m in the fixed femur is denoted by $(\bar{r}_m)_f$ in the base $(\hat{i}, \hat{j})$. Here the subscripts t and f outside the parentheses imply 'tibia' and 'femur', respectively. The current length of the ligament is given by

$$L_m = \sqrt{[(\bar{r}_m)_f - \bar{r}_o - T(\bar{\rho}'_m)_t] \cdot [(\bar{r}_m)_f - \bar{r}_o - T(\bar{\rho}'_m)_t]}. \quad (13)$$

The unit vector, $\hat{\lambda}_m$, along the ligament m directed from the tibia to the femur is

$$\hat{\lambda}_m = \frac{1}{L_m} [(\bar{r}_m)_f - \bar{r}_o - T(\bar{\rho}'_m)_t]. \quad (14)$$

Thus the axial force in the ligament m , in its vectorial form, becomes

$$\bar{F}_m = F_m \hat{\lambda}_m \quad (15)$$

where F_m is given by equations (12).

The coordinates of the insertion points of the ligaments in both the tibia and the femur are determined from the available information in literature (Wang *et al.*, 1973; Crowninshield *et al.*, 1976) and close anatomical study of the knee joint. These coordinates on the tibia are denoted by x'_m and y'_m ; and those on the femur are denoted by x_m and y_m . The values used in the present work are summarized in Table 2. Obviously, these values are for a specific specimen used in our study.

Since the friction force between the tibia and the femur is neglected, the contact force will be in the direction of the normal to the surface at the point of

Table 1. Ligament stiffnesses

Ligament	K_m (N mm ⁻²)
LC	15
MC	15
AC	30
PC	35

Table 2. Coordinate values of the insertions and origins of the ligaments in meters

Ligament	Tibia		Femur	
	x'_m	y'_m	x_m	y_m
Medial collateral (MC)	0.008	0.163	-0.023	0.014
Lateral collateral (LC)	0.025	0.178	-0.025	0.019
Anterior cruciate (AC)	-0.005	0.213	-0.023	0.019
Posterior cruciate (PC)	0.025	0.208	-0.032	0.024

contact. The contact force $\bar{N}$ acting on the tibia is given by

$$\bar{N} = \gamma N \hat{n}_1 \quad \text{at} \quad x = x_c \quad (16)$$

where N is the magnitude of the contact force and

$$\gamma = \frac{(d^2 f_1 / dx^2)}{|d^2 f_1 / dx^2|} \quad \text{at} \quad x = x_c \quad (17)$$

γ is either +1 or -1 and it ensures the correct direction of the contact force acting on the tibia.

The final set of forces entering the equations of motion are the external forces (Fig. 2) whose resultant is given as:

$$\bar{F}_e = (F_e)_x \hat{i} + (F_e)_y \hat{j}, \quad (18a)$$

along with the external moment resultant

$$\bar{M}_e = M_e \hat{k}. \quad (18b)$$

Equations of motion

The equations governing the forced motion of the tibia are as follows:

$$(F_e)_x + \gamma N (n_1)_x + \sum_{m=1}^4 F_m (\lambda_m)_x = M \ddot{x}_o \quad (19a)$$

$$(F_e)_y + \gamma N (n_1)_y + \sum_{m=1}^4 F_m (\lambda_m)_y = M \ddot{y}_o \quad (19b)$$

$$M_e + (T \bar{\rho}'_c) (\gamma N \bar{n}_1) + \sum_{m=1}^4 (T \bar{\rho}'_m) (F_m \bar{\lambda}_m) = I_z \ddot{\alpha}, \quad (19c)$$

in which the subscripts x and y denote the components of the related quantities in x and y -directions. The mass of tibia is denoted by M and the dots denote derivatives with respect to time t . The mass moment of inertia of the tibia about the z -axis is I_z and $\ddot{\alpha}$ designates the angular acceleration of the tibia. The problem description is completed by assigning the initial conditions which are

$$\dot{x}_o = \dot{y}_o = \dot{\alpha} = 0, \quad (20)$$

along with specified values for x_o , y_o and α at $t = 0$. The numerical procedure employed in the solution of the governing equations is described in the following section.

NUMERICAL PROCEDURE

The governing equations of the initial value problem at hand are the three equations of motion (19), the contact condition (10) and two geometric compatibility conditions (7). The main unknowns of the problem are x_o , y_o , α , N , x_c and x'_c . The problem is, thus, reduced to the solution of a set of simultaneous nonlinear differential and algebraic equations.

The first step in arriving at a numerical solution of these equations is the replacement of the time derivatives with a temporal operator; in the present work, the Newmark operators (Bathe and Wilson, 1976) are chosen for this purpose. For instance, $\ddot{x}_o$ is expressed in the following form:

$$\ddot{x}_o = \frac{4}{(\Delta t)^2} (x_o^t - x_o^{t-\Delta t}) - \frac{4}{\Delta t} \dot{x}_o^{t-\Delta t} - \ddot{x}_o^{t-\Delta t}, \quad (21a)$$

$$\dot{x}_o = \dot{x}_o^{t-\Delta t} + \frac{\Delta t}{2} \ddot{x}_o^{t-\Delta t} + \frac{\Delta t}{2} \ddot{x}_o^t, \quad (21b)$$

in which Δt is the time increment and the superscripts refer to the time stations. Similar expressions are used for $\ddot{y}_o$ and $\ddot{\alpha}$. In the application of equations (21), the conditions at the previous time station ($t - \Delta t$) are, of course, assumed to be known.

After the time derivatives in equation (19) are replaced with the temporal operators defined, the governing equations take the form of a set of nonlinear algebraic equations. The solution of these equations is complicated by the fact that iteration or perturbation methods must be used. In this work, the Newton-Raphson (Kao, 1974) iteration process is used for the solution. To linearize the resulting set of simultaneous algebraic equations we assume

$$^k X_o^t = {}^{k-1} X_o^t + \Delta X_o \quad (22)$$

and similar expressions for the other variables are written. Here, the right superscripts denote the time station under consideration and the left superscripts denote the iteration number. At each iteration k the values of the variables at the previous ($k - 1$)st iteration are assumed to be known. The delta quantities denote incremental values. Equation (22) and the corresponding ones for the other variables are substituted into the governing nonlinear algebraic equations and the higher order terms in the delta quantities are dropped.

The set of n simultaneous algebraic (now linearized) equations can be put into the following matrix form

$$[K]\{\Delta\} = \{D\} \quad (23)$$

where $[K]$ is an $n \times n$ coefficient matrix, $\{\Delta\}$ is a vector of incremental quantities and $\{D\}$ is a vector of known values. Although the true value of n is 6, because of the length and complexity of the equations new variables were defined, thus, during actual computation process n became 22.

The iteration process at a fixed time station continues until the delta quantities of all the variables become negligibly small. In the present work, a solution is accepted and iteration process is terminated when the delta quantities become less than or equal to 0.01 % of the previous values of the corresponding variables. The converged solution of each variable is then used as the initial value for the next time step and the process is repeated for consecutive time steps.

The only problem that the Newton-Raphson process may present in the solution of dynamic problems is due to the fact that the period of the forced motion of the system may turn out to be quite short. In this case it becomes necessary to use very small time steps; otherwise, a significantly large number of iterations is required for convergence. In our application of the method, only 5-6 iterations were necessary for convergence most of the time. But there were also cases where more iterations were required for convergence. This was especially true at instants at which there was a sudden sharp change in the response of the tibia. Such behavior manifested itself usually when the tibia started moving in the opposite direction due to the fact that the pulling force of a ligament(s) became the governing force of the problem. Shorter time steps required fewer iterations, as expected, even at the situation described. The time increment used in the present work is $\Delta t = 0.0001$ s.

NUMERICAL RESULTS AND DISCUSSION

The numerical results to be presented are only for an external force acting on the tibia without the presence of an external moment. It is assumed that the force is always perpendicular to the longitudinal axis of the tibia (y' -axis) and passes through its center of mass. Let this force be denoted by $F_e(t)$. A parametric study of the effect of various combinations of moment and force acting simultaneously on the response of the knee joint may prove to be rewarding. However, for the present work we will only consider an external force and believe that this will be sufficient to illustrate the capabilities of the model. The effect of the shape of the forcing function on the knee joint response will be studied by considering the following two functions for $F_e(t)$:

$$F_e(t) = A[H(t) - H(t - t_0)] \quad (24)$$

which is a rectangular pulse of duration t_0 , and

amplitude A ; and

$$F_e(t) = Ae^{-4.73(t/t_0)^2} \sin\left(\frac{\pi t}{t_0}\right), \quad (25)$$

which is an exponentially decaying sinusoidal pulse of duration t_0 , and amplitude A . A dynamic loading in the form of equation (24) is extremely difficult to simulate experimentally; however, the study of these two functions will hopefully be helpful in understanding the effect of rise time of the dynamic load on the joint response. Equation (25) is a more realistic forcing function and it has been previously used as a typical representation of the dynamic load in head impact analysis (Engin and Akkas, 1978).

The following are obtained as a function of time from the computer program developed for this study; the coordinates of x_o , y_o of the center of mass of tibia; the flexion angle α ; the coordinates x_c and x'_c of the contact point in (x, y) and (x', y') coordinate systems respectively; the magnitude, N , of the contact force; the elongations of the ligaments and the ligament forces, F_m .

The initial values for x_o , y_o and α are obtained by specifying the location of the starting contact point. Here, the following values are used for the coordinates of the contact point at $t = 0$:

$$x_c = -4.2 \text{ cm}, \quad x'_c = 2.5 \text{ cm}$$

which yields

$$x_o = -20.16 \text{ cm}, \quad y_o = 17.49 \text{ cm}, \quad \text{and} \quad \alpha = 234.79^\circ$$

This angle of rotation α , corresponds to a flexion angle of 54.79° for which the ligaments of the knee joint are in a relatively relaxed condition. It was reported as early as 1907 by Pringle that the position of maximum relaxation of the knee joint ligaments was approximately halfway between full flexion and full extension of the knee joint.

The effect of pulse duration on the response of the knee joint motion is studied by taking $t_0 = 0.05, 0.10$ and 0.15 sec for both rectangular and exponentially decaying sinusoidal pulses. The effect of pulse amplitude, A , is also examined by taking $A = 20\text{N}, 60\text{N}, 100\text{N}, 140\text{N}$ and 180N for both types of pulses. In principle, it is possible to plot and cross-plot the results in numerous ways. However, because of the limitations on the space available, in the following a few typical results will be presented. More extensive results can be found in Moeinzadeh (1981).

Ligament forces as functions of flexion angle of the knee joint for the two previously described forcing functions are presented in Figs. 3(a) and 3(b). Results indicate that when the knee joint is extended by a dynamic application of a pulse on the tibia, lateral collateral, medial collateral and anterior cruciate ligaments are elongated while the posterior cruciate ligament is shortened. The load carried by the anterior cruciate ligament is substantially higher than those of the lateral collateral and medial collateral ligaments.

The variation of the lengths of ligaments and the

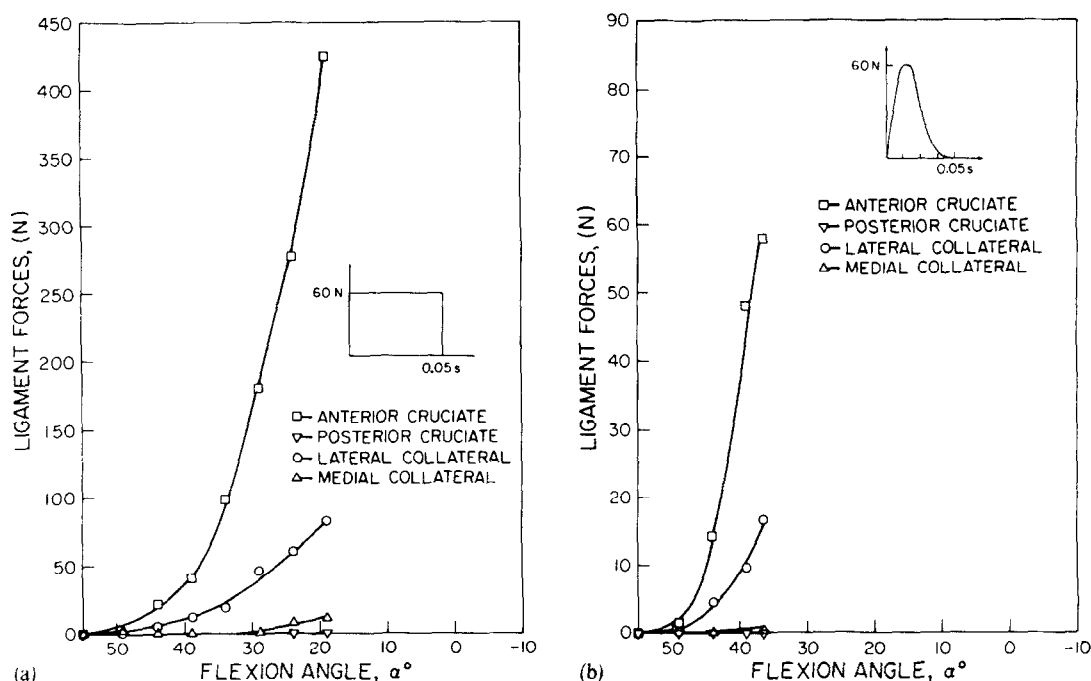


Fig. 3. Ligament forces as functions of flexion angle, for an externally applied (a) rectangular and (b) exponentially decaying sinusoidal pulses of 60 amplitude and 0.05 s duration.

forces carried by them during normal knee motion has been the subject of various studies reported in the literature and in these studies several different opinions and conclusions have been expressed in regard to the biomechanical role and function of various ligaments of the knee joint. The function of the anterior cruciate as depicted in the dynamic knee-model developed in this study is to resist anterior displacement of the tibia. This function is in general agreement with the experimental and clinical studies of Kennedy and Fowler (1971); Girgis *et al.* (1975); Van Dijk *et al.* (1979); and quasi-static model analyses of Crowninshield *et al.* (1976); and Wismans *et al.* (1980). The role of the posterior cruciate as predicted by the model is to resist posterior displacement of the tibia. This function of the posterior cruciate ligament is in agreement with the experimental studies of Kennedy and Grainger (1967); Edwards *et al.* (1970); Girgis *et al.* (1975); Crowninshield *et al.* (1976) and Wismans *et al.* (1980).

The present dynamic model also predicts that the medial and lateral collateral ligaments offer very little resistance in the flexion-extension motion of the knee joint. The major role of these ligaments is to offer varus-valgus and partial internal-external rotational stability. The model shows that as the knee joint is extended under influence of a dynamic load the lateral collateral and medial collateral ligaments elongate at different magnitudes. This prediction is in general agreement with the experimental results of Smillie (1970); Edwards *et al.* (1970); and Wang *et al.* (1973).

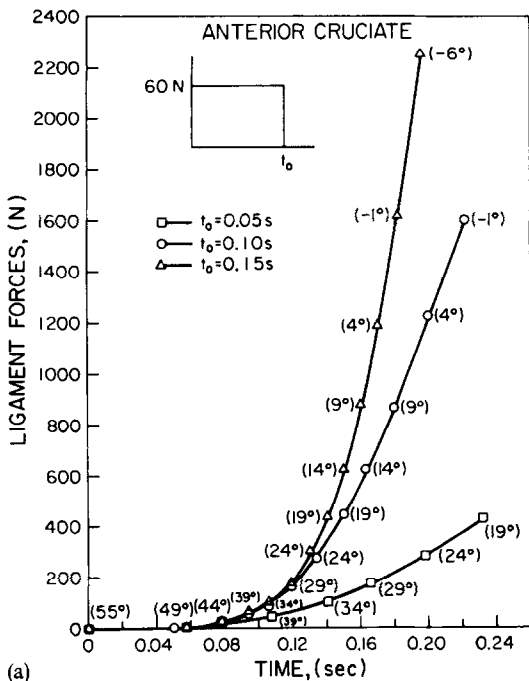
The model shows good agreement with the quasi-static experimental investigations reported in the literature.

It is important to note that the dynamic model presented in this study is an idealized representation of a very complex anatomical structure; thus, static experimental studies may not support some of the predictions of the model. Additional disagreements may also be due to approximate locations of the attachment sites of the ligaments in particular, and two-dimensional nature of the model in general.

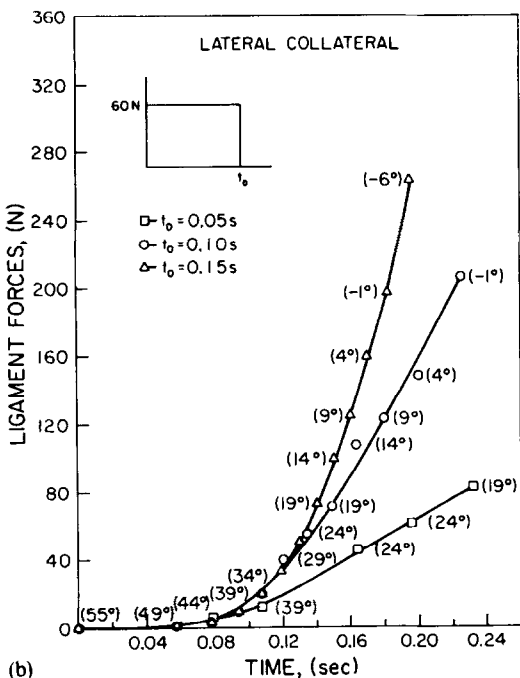
In Figs. 4(a) and 4(b), a few representative plots of forces in the anterior cruciate and lateral collateral ligaments are plotted as a function of time for two different forcing functions with varying pulse durations. Although not presented here, similar curves may be obtained for other pulse durations and pulse magnitudes. Generally, for a fixed amplitude, the shorter the pulse duration, the sooner the tibia reaches its turning point (i.e. direction of motion reverses) and for a given pulse duration, the smaller the amplitude, the sooner the turning point is reached. In Figs. 3-5 the values in parentheses indicate the flexion angles at the corresponding times. Note that, for illustrative purposes up to 6° of hyperextension was allowed. Generally, one expects only 1-3° of hyperextension to be anatomically tolerable beyond which joint failure becomes unavoidable.

In Fig. 5, contact forces as a function of time are plotted. These forces are in response to the different forcing functions with varying amplitudes and pulse durations. Note that the magnitudes of the ligament and the corresponding contact forces in response to a particular forcing function are comparable.

Femoral and tibial contact point locations as a function of flexion angle are plotted in Fig. 6. In these



(a)



(b)

Fig. 4. (a) Anterior cruciate and (b) lateral collateral ligament forces as a function of time for externally applied rectangular pulses of 60 N amplitude and durations of 0.05, 0.10 and 0.15 s.

figures the values in the parentheses indicate the total elapsed time of the motion since its initiation. As the flexion angle decreases, it can be seen that the curves representing femoral contact points have a steadily increasing positive slope, while the curves of tibial contact points change slope from positive to negative or vice-versa at various flexion positions. This phenom-

enon may be explained by the combined rolling and sliding motion of the tibia on the femur; that is, positive and zero slopes of the curve representing the tibial contact points can be interpreted as corresponding to the sliding motion, while negative slopes indicate the rolling motion of tibia on femur. Generally, the curves of tibial contact points have predominately negative slopes toward the end of the extension motion indicating that in this part of the motion rolling is the essential component. These results are in general agreement with the work of Walker and Hajek (1972); Kettelkamp and Jacobs (1972); and Wismans *et al.* (1980). Finally, for illustrative purposes, with the aid of Versatec plotter, continuous plots of x_o and y_o (coordinates of the center of mass of the tibia); x_c and x'_c (femoral and tibial contact points, respectively) and flexion angle, α , as a function of time are plotted in Fig. 7.

SUMMARY AND CONCLUDING REMARKS

The research work presented in this paper can be summarized in the following paragraphs:

1. A two-dimensional mathematical dynamic model of the knee joint has been developed to describe the relative dynamic motions between femur and tibia and the forces in the joint. The model includes the curvatures effects of the articular surfaces as well as the nonlinear behavior of the major ligaments of the joint.

2. A rectangular and an exponentially decaying sinusoidal pulse of duration, t_0 , and amplitude, A , were applied to the tibia and the numerical results from this model were presented to illustrate the effects of duration and shape of the dynamically applied loads on the response of the joint. Special attention has been given to the ligament and contact forces, the location of contact points and anterior-posterior displacements. The results were compared with the available experimental data from the literature to establish the validity of the model.

3. The model predicts that when the knee joint is extended, by a dynamic application of a pulse on the tibia, lateral collateral, medial collateral and anterior cruciate ligaments are elongated while the posterior cruciate ligament is shortened. The load carried by the anterior cruciate ligament is substantially higher than those of the lateral collateral and medial collateral ligaments. However, because of the planar nature of the model, the ligament functions predicted by this model may not be quite reflective of the actual function of the knee joint ligaments and should be viewed only as an approximation.

With this model the influence of the variations of initial strain of the ligaments and their attachment sites (i.e. insertions and origins) on the response of the model may be determined and a parametric study addressing to these points may reveal the sensitivity of the model to the variations of the coordinates of the insertions and origins of the ligaments.

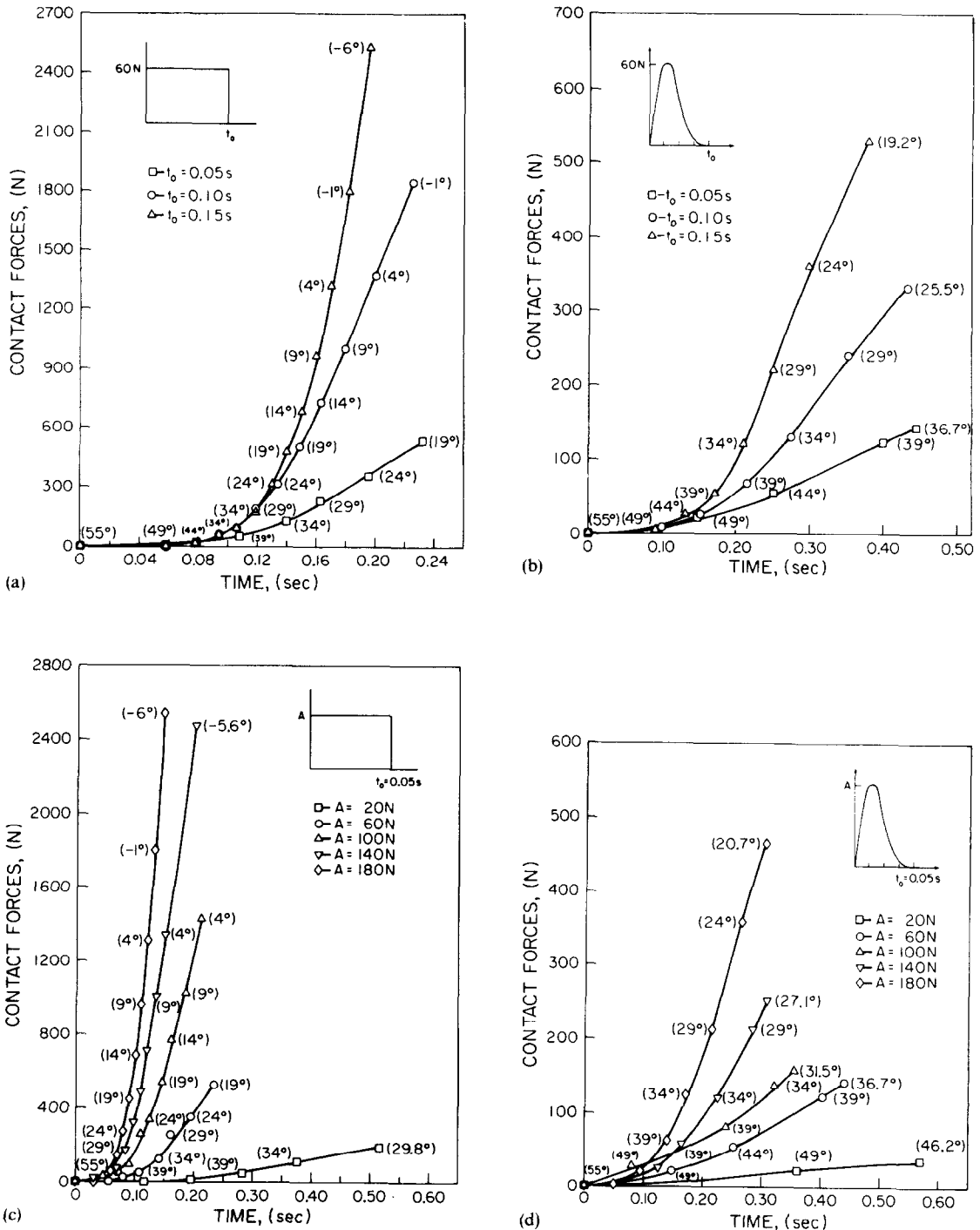


Fig. 5. Contact forces as a function of time for an externally applied (a) rectangular pulse of 60 N amplitude and durations of 0.05, 0.10 and 0.15 s, (b) exponentially decaying sinusoidal pulse of 60 N amplitude and durations of 0.05, 0.10 and 0.15 seconds, (c) rectangular pulses of 20 N, 60 N, 100 N, 140 N and 180 N amplitude and 0.05 s duration and (d) exponentially decaying sinusoidal pulses of 20 N, 60 N, 100 N, 140 N and 180 N amplitude and 0.05 s duration.

In a similar way the mathematical models of the other major articulating joints can be developed. However, a special attention should be given in modelling of the ligaments. Some ligaments, particularly the thick band or large cord-like ligaments, have

complex behavior, with various portions behaving differently under given conditions or configurations. These are described in the literature as, in the case of the broad ligaments of the hip joint, having anterior and posterior fibers, or medial and lateral components.

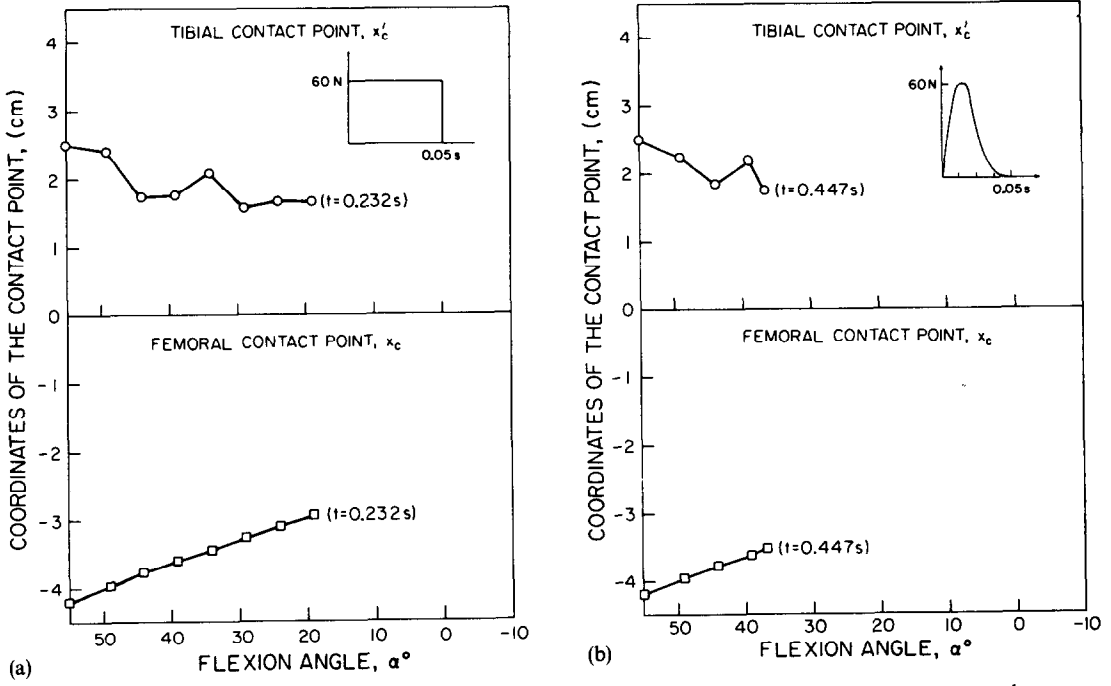


Fig. 6. Femoral and tibial contact points as a function of flexion angle for externally applied (a) rectangular and (b) exponentially decaying sinusoidal pulses of 60 N amplitude and 0.05 s duration.

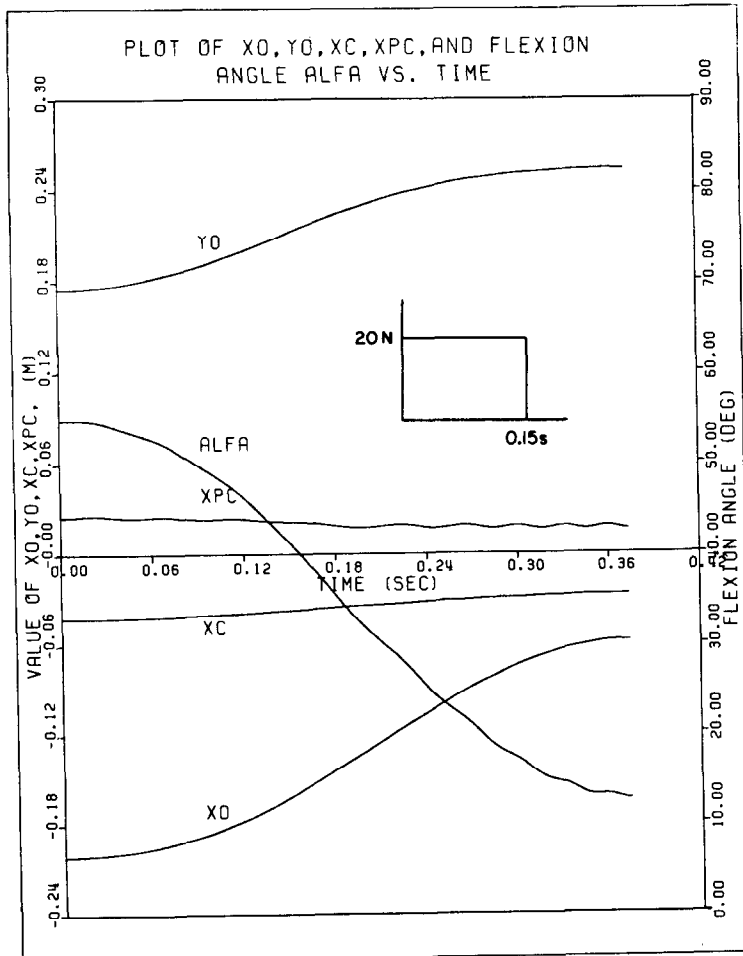


Fig. 7(a)

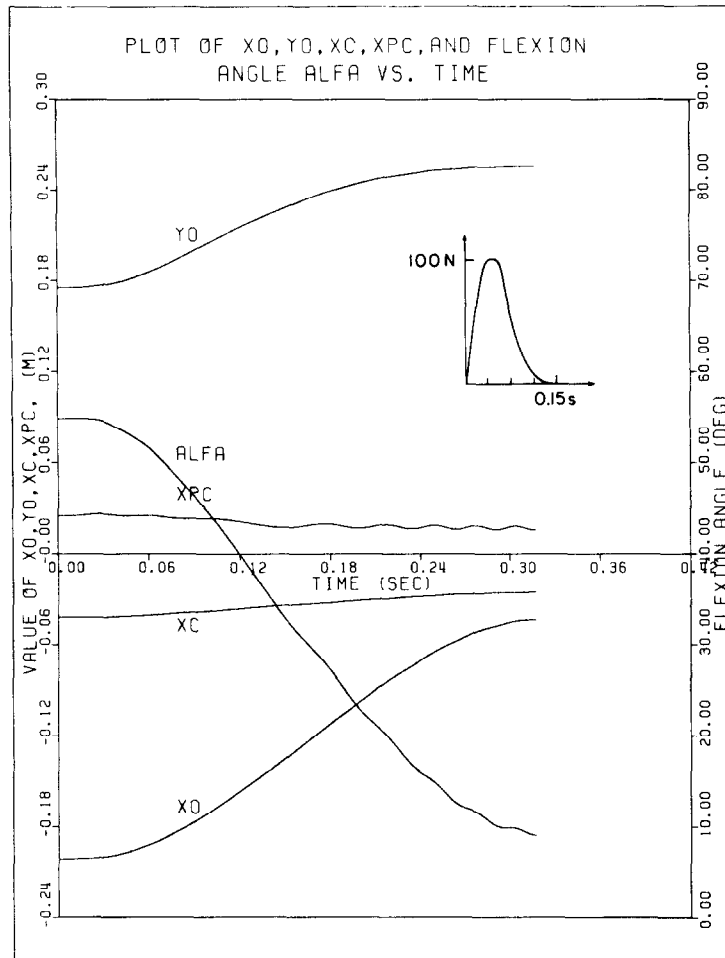


Fig. 7. Continuous plots of tibial center of mass coordinates (XO , YO), femoral (XC) and tibial (XPC) contact points and the flexion angle ($ALFA$) as functions of time for an externally applied (a) rectangular pulse of 20 N amplitude and 0.15 s duration and (b) exponentially decaying sinusoidal pulse of 100 N amplitude and 0.15 s duration.

The mathematical model should include additional elastic elements to reflect the contributions of various fibers of the ligaments. Unfortunately proper experimental data to determine the constitutive behavior of these thick band or broad ligaments do not exist.

Finally, in advancing the understanding of ligament and articular surface functions of human joints under dynamic external loading conditions, the method of joint modelling presented in this paper and the numerical procedure utilized in the solution of equations of the two-dimensional dynamic knee joint model may be extended to the solution of equations of three-dimensional dynamic model of an articulating joint. In fact, the three-dimensional articulating surface equations for the elbow, hip, knee and ankle joints as well as the origins and insertions of the major ligaments of these joints have been already provided in Moeinzadeh (1981). Development and results obtained from the analysis of the three-dimensional dynamic joint model will be presented in a future paper.

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